

SETTING UP NETWORKING SIMULATION TOOLS AND INSTALLING WIRESHARK FOR NETWORK TRAFFIC ANALYSIS

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ABSTRACT

Network traffic analysis is a fundamental skill in cybersecurity, network administration, and digital forensics. This project focused on the installation, configuration, and utilization of Wireshark for capturing and analyzing network traffic within a home network environment. The project involved identifying active network interfaces, capturing live traffic, examining common network protocols, verifying network connectivity, and documenting implemented security controls. The analysis revealed the presence of DNS, TCP, ICMP, UDP, and TLS communications, demonstrating the practical application of packet analysis in monitoring and troubleshooting network activities. The project provided hands-on experience with network monitoring techniques commonly used by cybersecurity professionals, network administrators, and security analysts.

1. INTRODUCTION

Modern computer networks form the backbone of communication systems used by individuals, businesses, and governments. As network usage continues to increase, cybersecurity professionals require tools that enable them to monitor, analyze, and secure network communications.

Wireshark is one of the most widely used network protocol analyzers in the world. It allows users to capture, inspect, and analyze packets traversing a network in real time. By examining

captured traffic, security analysts can identify communication patterns, troubleshoot connectivity issues, investigate incidents, and detect suspicious activities.

This project focused on establishing a practical network monitoring environment using Wireshark. The project included software installation, network traffic capture, protocol analysis, connectivity testing, and documentation of implemented security controls.

2. PROJECT OVERVIEW

This project introduced the concepts of network simulation and packet analysis through the installation and use of Wireshark.

The project aimed to provide practical experience in:

- Network monitoring
- Packet capture
- Protocol analysis
- Connectivity verification
- Security assessment
- Troubleshooting network communications

The project was conducted within a home wireless network environment.

3. PROJECT OBJECTIVES

The objectives of this project were to:

1. Install and configure Wireshark.
 2. Install and configure Npcap packet capture drivers.
 3. Identify active network interfaces.
 4. Capture live network traffic.
 5. Analyze common network protocols.
 6. Verify connectivity between devices.
 7. Document network architecture.
 8. Examine implemented security controls.
 9. Gain practical experience with cybersecurity monitoring tools.
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4. PROJECT SCOPE

This project falls under the area of Network Monitoring and Traffic Analysis.

The scope includes:

- Packet capture
- Protocol identification
- Network connectivity testing
- Host-based security verification
- Wireless security assessment
- Network troubleshooting

The project does not involve offensive security testing or penetration testing activities.

5. TOOLS AND SOFTWARE USED

Table 1: Tools Used

Tool: Wireshark 4.6.6 Purpose: Packet Capture and Analysis

Tool: Npcap Purpose: Packet Capture Driver

Tool: Windows 11 Purpose: Host Operating System

Tool: Wi-Fi Network Adapter Purpose: Network Traffic Source

Tool: Android Smartphone Purpose: Additional Network Device

6. ASSETS, THREATS, AND VULNERABILITIES

6.1 Assets Identified

The following assets were identified:

- Windows Laptop
- Home Router
- Android Smartphone
- Network Traffic
- Internet Connection
- Wireshark Capture Files

6.2 Potential Threats

• Packet Sniffing • Unauthorized Access • Malware Communications • Data Interception • Network Reconnaissance

6.3 Potential Vulnerabilities

• Weak Wireless Security • Unmonitored Network Traffic • Misconfigured Systems • Unauthorized Devices

7. WIRESHARK INSTALLATION AND CONFIGURATION

Wireshark was downloaded from the official Wireshark website and installed successfully.

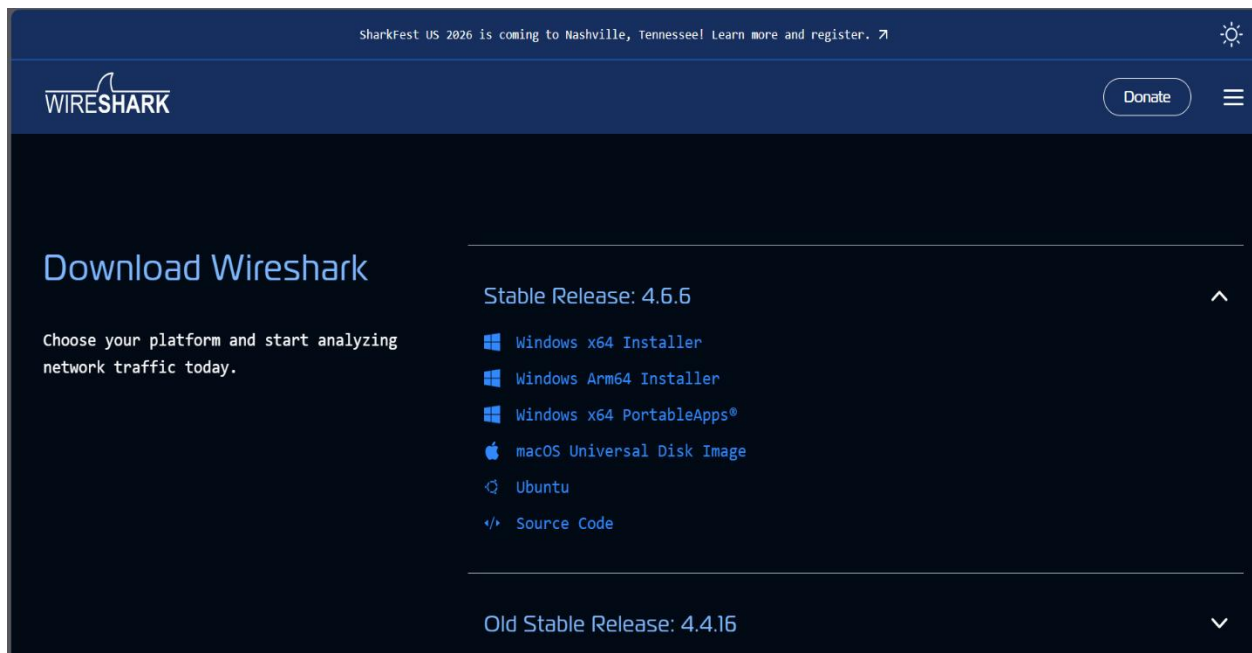
The installation included:

• Wireshark Network Analyzer • Npcap Packet Capture Driver • Protocol Analysis Components

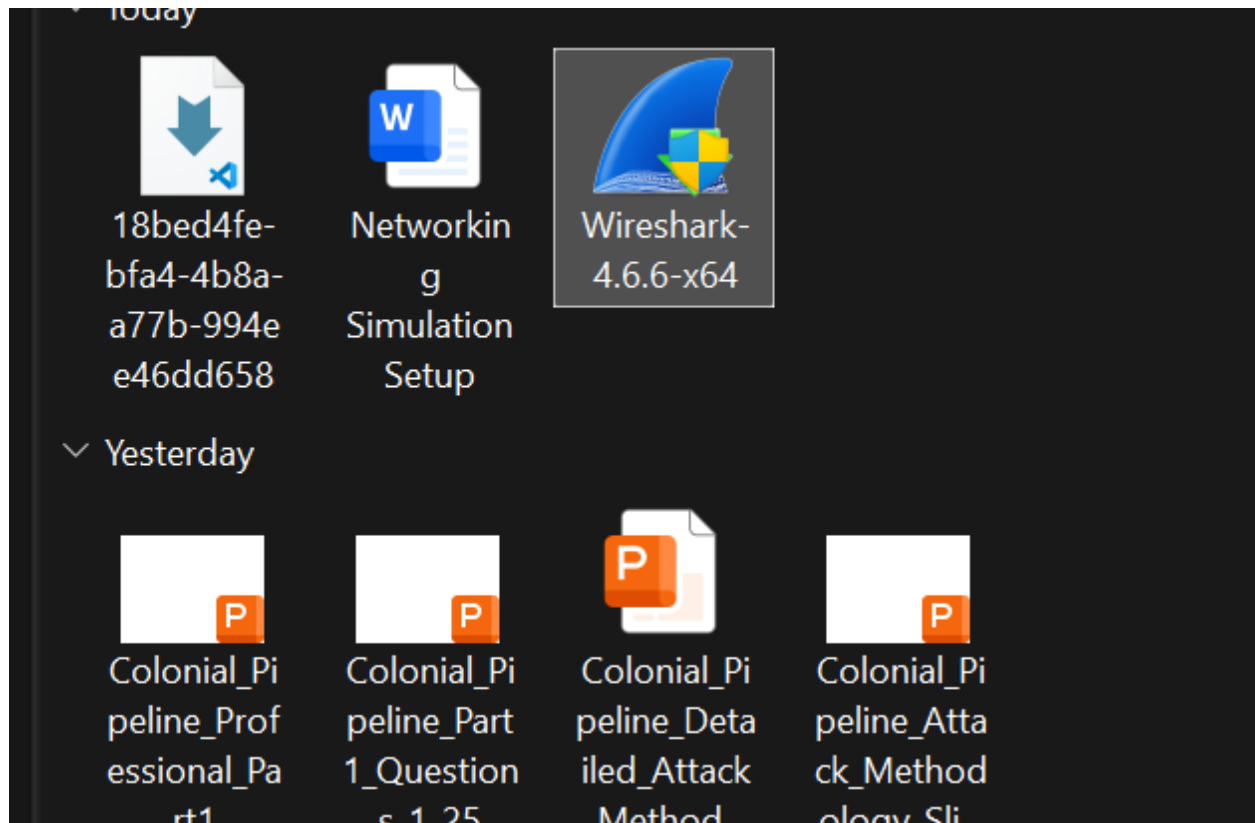
Installation Steps:

1. Downloaded Wireshark from the official website.
2. Executed the installer package.
3. Accepted the license agreement.
4. Installed Npcap packet capture drivers.
5. Completed software installation.
6. Verified successful launch.

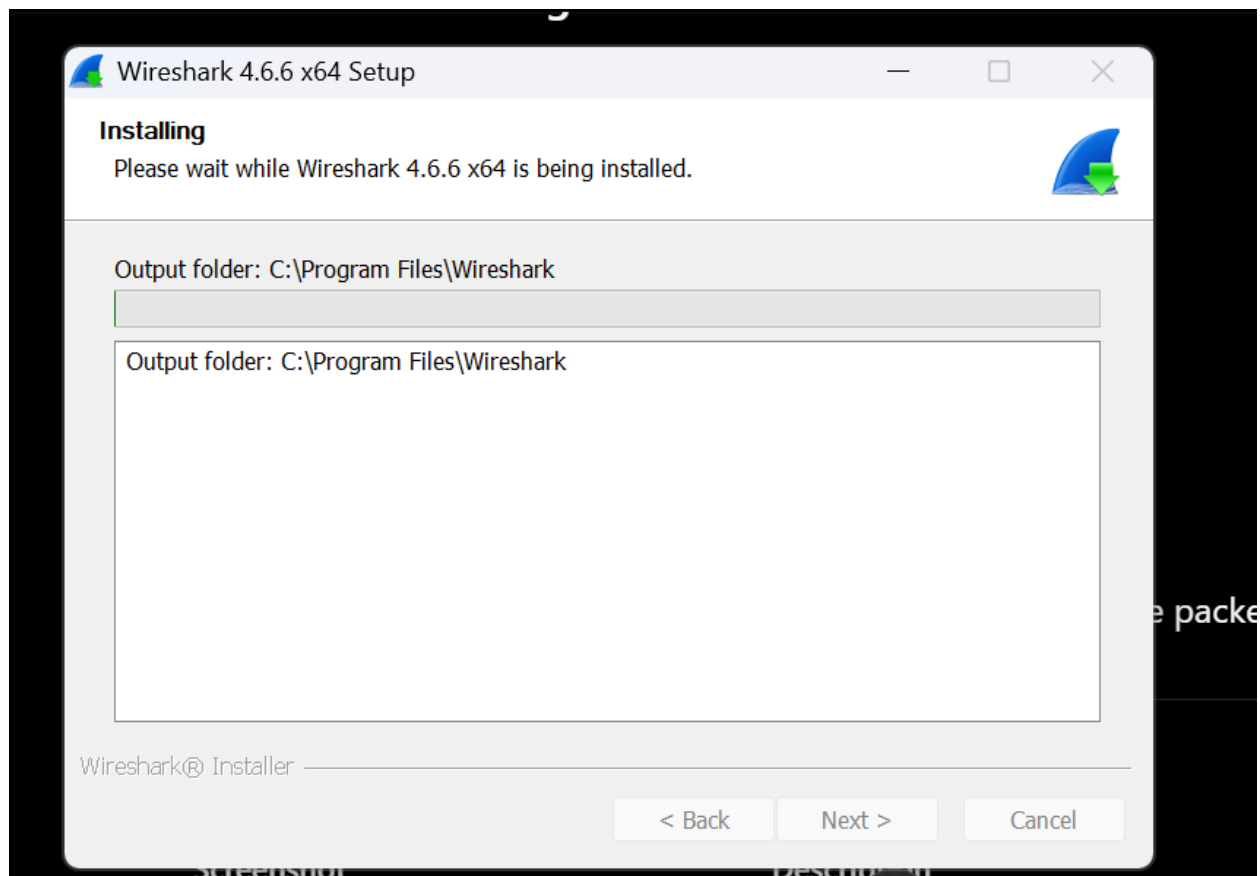
Wireshark Download Page



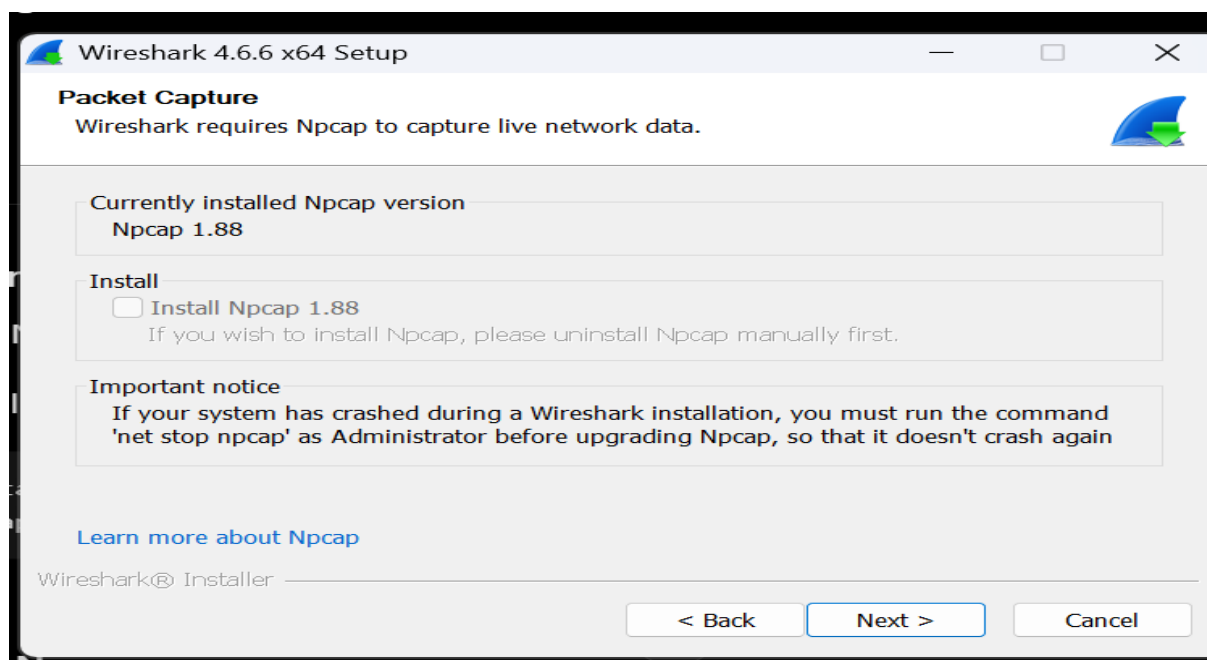
Downloaded Wireshark Installer



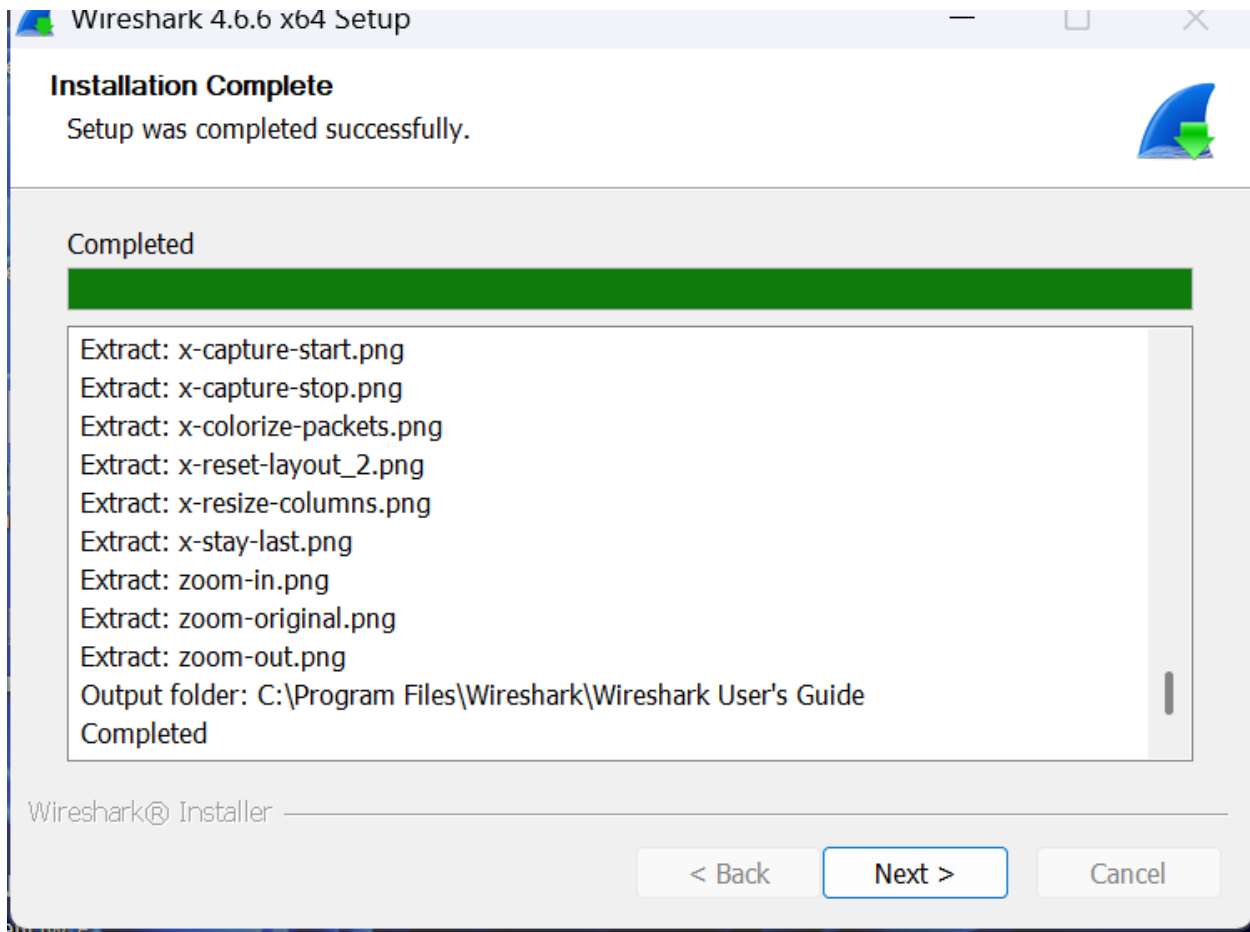
Wireshark Installation Wizard



Npcap Installation



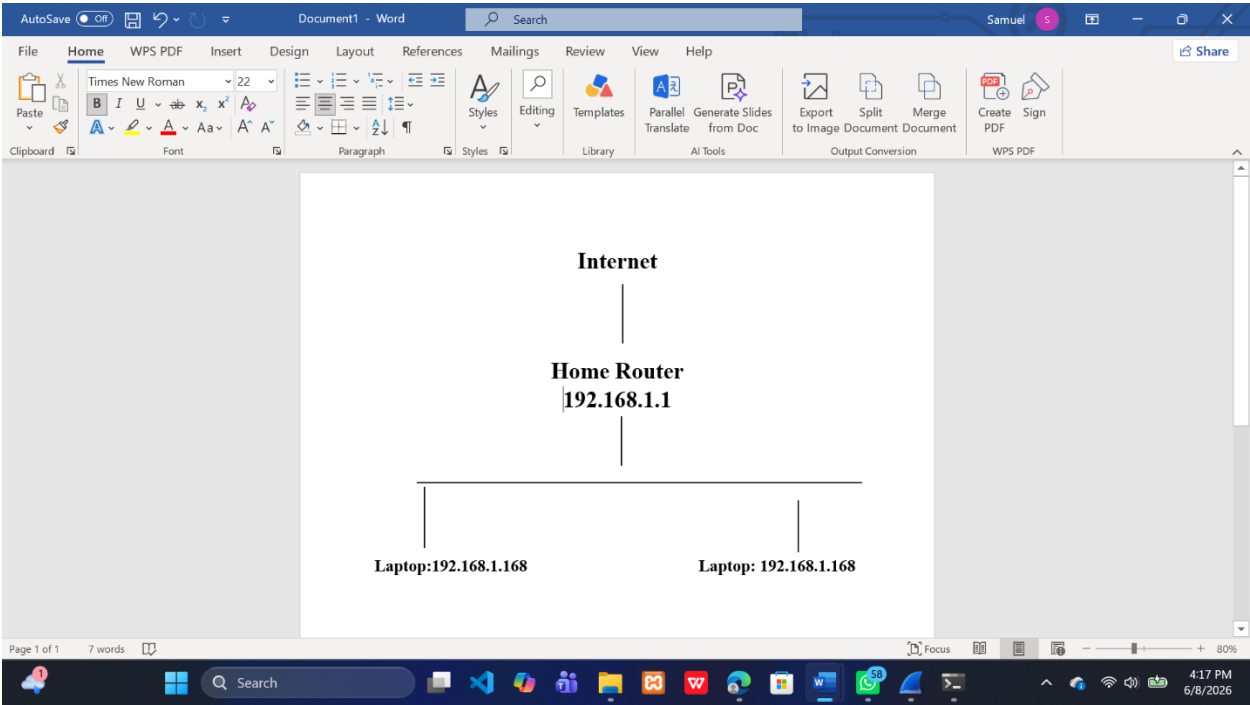
Installation Completed Successfully



8. NETWORK TOPOLOGY AND ARCHITECTURE

The monitoring environment consisted of a wireless home network.

Figure 1: Home Network Topology



The router acted as the default gateway and provided network connectivity to all devices.

Insert Figure 1: Home Network Topology Diagram

9. NETWORK DEVICE CONFIGURATION

Table 2: Connected Devices

Device: Home Router IP Address: 192.168.1.1 Function: Default Gateway

Device: Laptop IP Address: 192.168.1.168 Function: Packet Analysis Workstation

Device: Android Phone IP Address: 192.168.1.136 Function: Network Client Device

Network Configuration:

IPv4 Network: 192.168.1.0/24

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.1

Laptop IP Configuration (ipconfig)

```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26100.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : home

Wireless LAN adapter Local Area Connection* 11:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 12:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::ccd3:2f66:39d:e049%15
    IPv4 Address. . . . . : 192.168.116.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2f41:72f4:8533:84c4%5
    IPv4 Address. . . . . : 192.168.190.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi 2:

    Connection-specific DNS Suffix  . : home
    Link-local IPv6 Address . . . . . : fe80::961:6822:1061:a70c%24
    IPv4 Address. . . . . : 192.168.1.168
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\DELL>
```

Android Network Configuration

Randomized MAC address	
ce:cc:cc:46:4e:de	
IP address	192.168.1.136
Gateway	192.168.1.1
Subnet mask	255.255.255.0
DNS	192.168.1.1
Transmit link speed	433 Mbps

10.CONNECTIVITY TESTING

Connectivity testing was performed to verify communication between devices.

Command Used:

ping 192.168.1.1

Results:

Packets Sent = 4

Packets Received = 4

Packet Loss = 0%

Average Response Time = 0ms

Interpretation:

The successful ping responses confirmed that the laptop could communicate successfully with the router.

Ping Connectivity Test

```
C:\Users\DELL>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\DELL>
```

11.ARP TABLE ANALYSIS

Address Resolution Protocol (ARP) was used to identify devices communicating on the local network.

Command Used:

arp -a

Purpose:

- Identify local devices
- Map IP addresses to MAC addresses
- Verify local communications

Findings:

- Router detected successfully
- Network communications verified
- Device addressing confirmed

Cybersecurity Importance:

ARP analysis assists in:

- Device discovery
- Network troubleshooting
- Detection of unauthorized devices
- Security investigations

ARP Table Output

```
Command Prompt
Default Gateway . . . . . : 192.168.1.1

C:\Users\DELL>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\DELL>arp -a

Interface: 192.168.190.1 --- 0x5
Internet Address      Physical Address      Type
192.168.190.254        00-50-56-f5-4a-ea    dynamic
192.168.190.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22             01-00-5e-00-00-16    static
224.0.0.251            01-00-5e-00-00-fb    static
224.0.0.252            01-00-5e-00-00-fc    static
239.255.255.250        01-00-5e-7f-ff-fa    static
255.255.255.255        ff-ff-ff-ff-ff-ff    static

Interface: 192.168.116.1 --- 0xf
Internet Address      Physical Address      Type
192.168.116.254        00-50-56-fe-0e-22    dynamic
192.168.116.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22             01-00-5e-00-00-16    static
224.0.0.251            01-00-5e-00-00-fb    static
224.0.0.252            01-00-5e-00-00-fc    static
239.255.255.250        01-00-5e-7f-ff-fa    static
255.255.255.255        ff-ff-ff-ff-ff-ff    static

Interface: 192.168.1.168 --- 0x18
Internet Address      Physical Address      Type
192.168.1.1            d8-42-f7-aa-40-59    dynamic
192.168.1.255          ff-ff-ff-ff-ff-ff    static
224.0.0.22             01-00-5e-00-00-16    static
224.0.0.251            01-00-5e-00-00-fb    static
224.0.0.252            01-00-5e-00-00-fc    static
239.255.255.250        01-00-5e-7f-ff-fa    static
255.255.255.255        ff-ff-ff-ff-ff-ff    static

C:\Users\DELL>
```

12.SECURITY CONTROLS IMPLEMENTATION

12.1 Windows Defender Firewall

Windows Defender Firewall was enabled on the system.

Observed Configuration:

Firewall State: Enabled

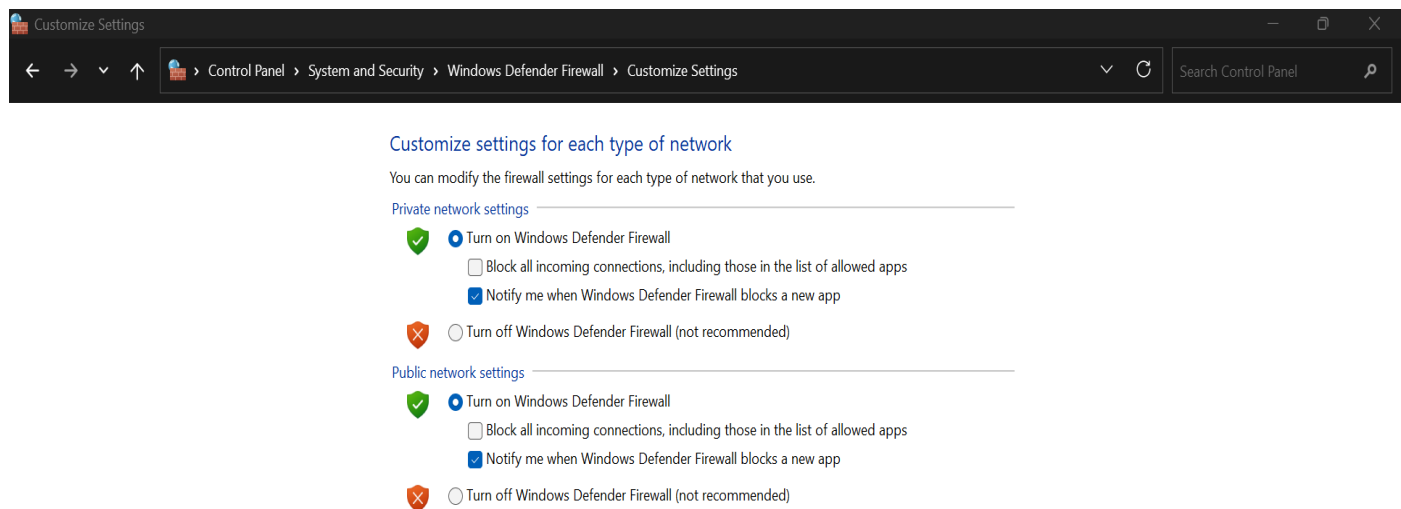
Public Network Protection: Enabled

Inbound Connections: Restricted

Benefits:

- Blocks unauthorized access
- Monitors network traffic
- Reduces attack surface
- Protects against malicious connections

Windows Defender Firewall Enabled



12.2 Wireless Security Assessment

Wireless security settings were examined.

Observed Security Configuration:

Security Type: WPA2-Personal

Encryption: AES

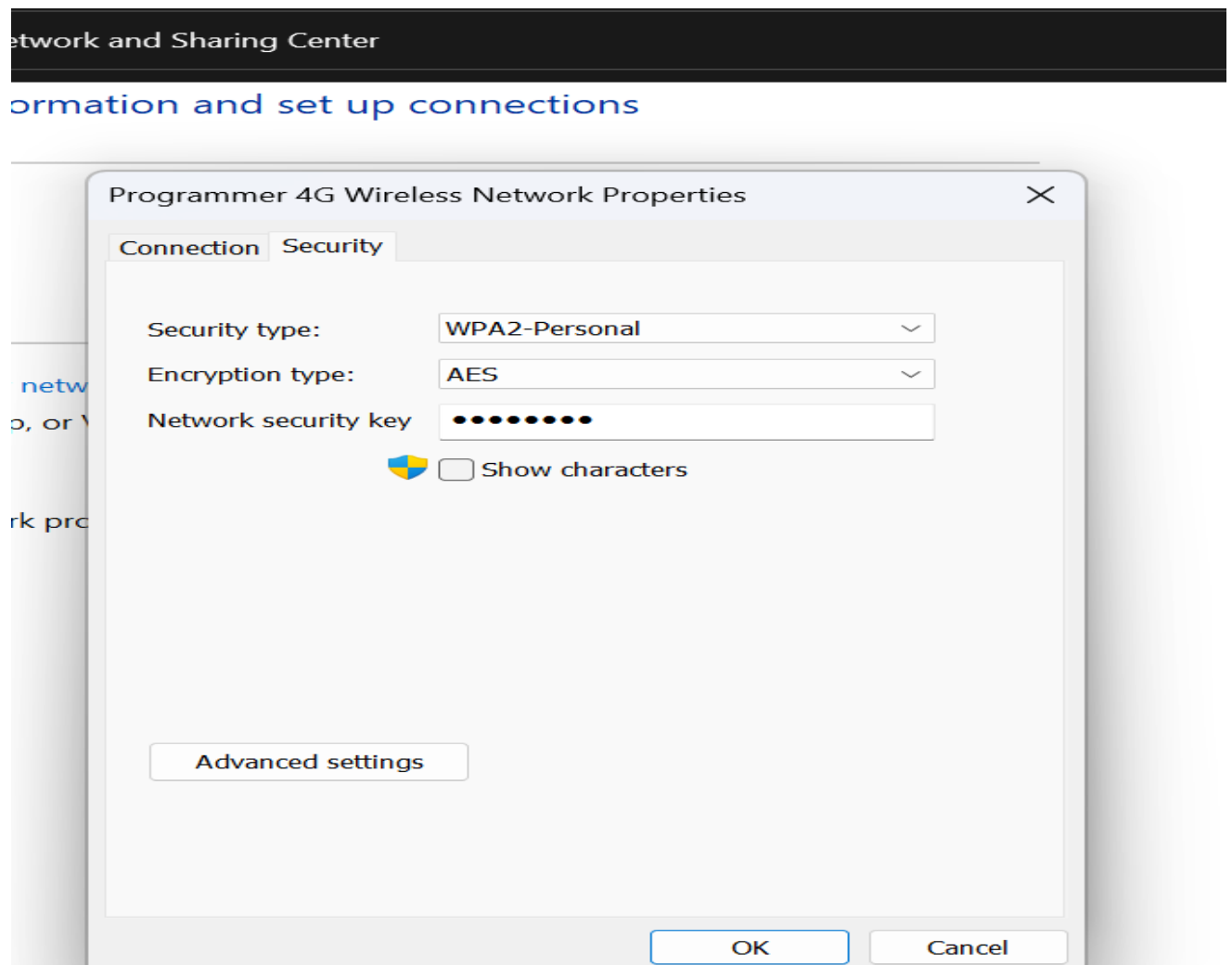
Analysis:

WPA2 with AES encryption protects wireless communications against unauthorized access and interception.

Benefits:

- Data confidentiality
- Network protection
- Secure wireless communications

Wireless Security Configuration



13.NETWORK INTERFACE DETECTION

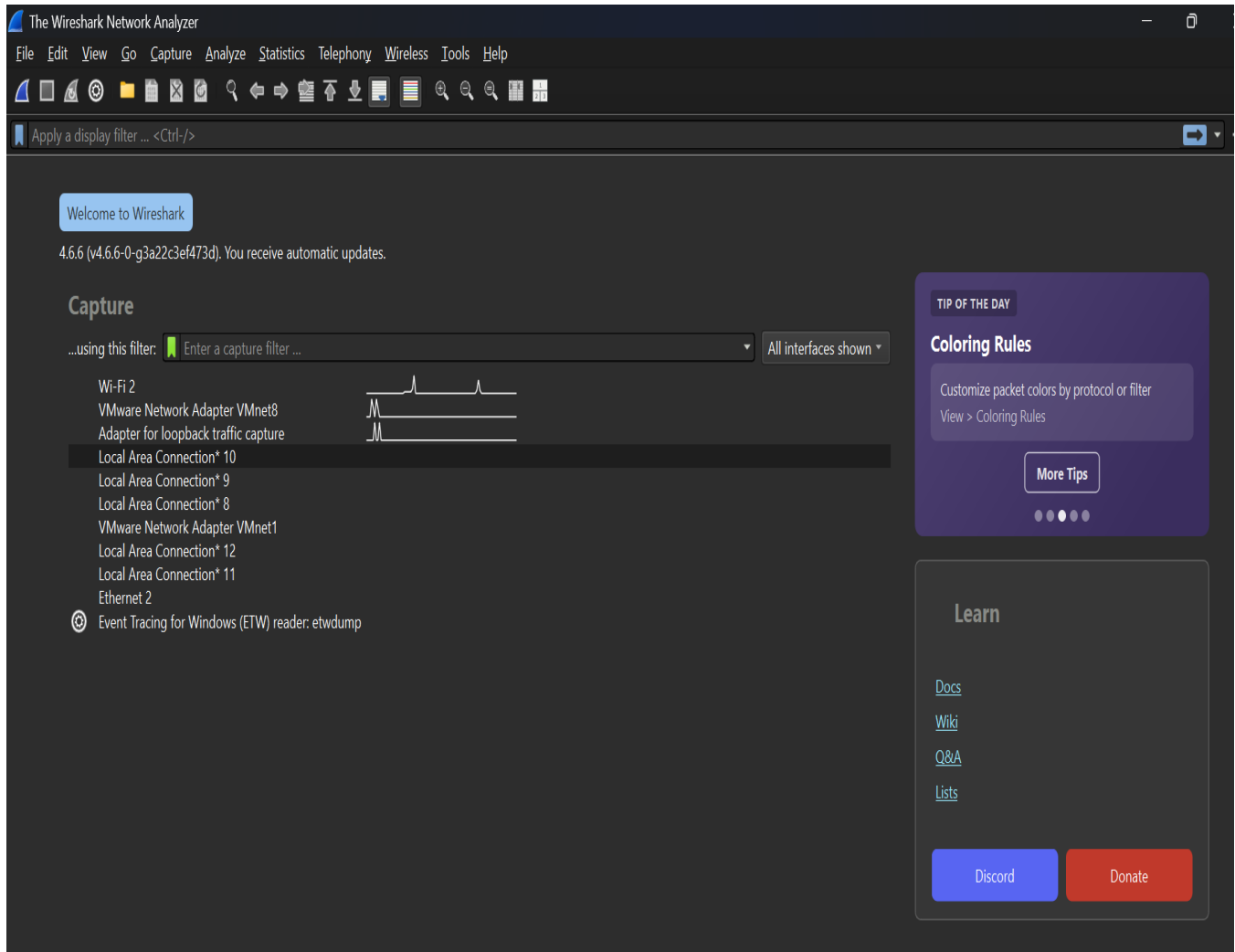
After launching Wireshark, several network interfaces were detected.

The active interface selected for packet capture was:

Wi-Fi 2

This interface showed active network traffic and was selected for analysis.

Available Network Interfaces



14.LIVE PACKET CAPTURE

A live packet capture session was initiated on the Wi-Fi 2 adapter.

Traffic was generated through:

- Web browsing • DNS lookups • HTTPS communications • Background network activities

Wireshark successfully captured and displayed packets in real time.

Live Packet Capture

Wireshark interface showing a packet capture from Wi-Fi 2. The packet list displays several packets, with packet 1162 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and User Datagram Protocol. The packet bytes pane displays the raw data of the packet in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1161	8.229222800	192.168.1.168	172.64.149.246	UDP	87	59271 → 443 Len=45
1162	9.996402100	192.168.1.168	172.64.149.246	UDP	1292	59271 → 443 Len=1250
1163	9.996840200	192.168.1.168	172.64.149.246	UDP	740	59271 → 443 Len=698
1164	10.158830700	172.64.149.246	192.168.1.168	UDP	66	443 → 59271 Len=24
1165	10.167654500	172.64.149.246	192.168.1.168	UDP	67	443 → 59271 Len=25
1166	10.206923400	172.64.149.246	192.168.1.168	UDP	805	443 → 59271 Len=763
1167	10.237305200	192.168.1.168	172.64.149.246	UDP	87	59271 → 443 Len=45
1168	11.499400200	192.168.1.168	74.248.73.245	TLSv1.2	105	Application Data
1169	11.654248700	74.248.73.245	192.168.1.168	TLSv1.2	94	Application Data
1170	11.699002600	192.168.1.168	74.248.73.245	TCP	54	55013 → 443 [ACK] Seq=52 Ack=41 Win=251 Len=0
1171	11.992809300	192.168.1.168	172.64.149.246	UDP	1292	59271 → 443 Len=1250
1172	11.993168000	192.168.1.168	172.64.149.246	UDP	740	59271 → 443 Len=698
1173	12.219483100	172.64.149.246	192.168.1.168	UDP	67	443 → 59271 Len=25
1174	12.272270300	172.64.149.246	192.168.1.168	UDP	803	443 → 59271 Len=761
1175	12.272841300	192.168.1.168	172.64.149.246	UDP	86	59271 → 443 Len=44

Frame 1: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on
 Ethernet II, Src: Intel_dc:60:1d (28:16:ad:dc:60:1d), Dst: TozedKangwei_aa:40:59 (d8:4
 Internet Protocol Version 4, Src: 192.168.1.168, Dst: 172.64.149.246
 User Datagram Protocol, Src Port: 59271, Dst Port: 443
 Data (1250 bytes)

```

0000  d8 42 f7 aa 40 59 28 16  ad dc 60 1d 08 00 45 00  -B @Y( . . . . .E
0010  04 fe dc 46 40 00 80 11  15 21 c0 a8 01 a8 ac 40  -...F...!...@
0020  95 f6 e7 87 01 bb 04 ea  27 7d 4a 01 02 90 6a 9e  -...}'J...j
0030  7d 4d bb db 00 86 0a 27  7d 5c de 43 ec 8c f9 a3  -}M...j'}VC...
0040  a1 24 d6 26 cc 29 0c 8c  ee ea 55 0f 45 51 02 80  -.$ & ) . . . . .U EQ
0050  9f 9a 31 fa 7e 81 92 98  d7 45 d5 15 ff 0d 34 00  -1 ~ . . . . .E . . . .
0060  65 f0 fa 2f 1c 4d 85 b4  e9 05 c4 b2 91 13 86 3e  -e / / M . . . . .>
0070  dc d3 ea fc df 9f f6 5d  7e 73 ea 9d 0a 0f 2c 56  -... ] ~ s . . . . .V
0080  bf 01 94 b3 99 0b 35 3f  ac df 10 50 06 25 76 32  -... 5? . . . . .P %v2
0090  fe 33 e2 3a c7 64 f2 73  a0 b9 88 9b 45 5c f3 24  -3 . : d s . . . . .E $
00a0  64 2e 19 60 08 3d 28 57  8d 86 f2 b9 a8 6d 4d 5e  d . . . . .(W . . . . .mM^
00b0  a6 c7 aa e9 d3 46 f6 5e  d1 5e d3 07 e9 11 d2 15  -...FoA . . . . .^
00c0  83 a0 9f d5 9b 31 22 2a  40 e0 fe d6 9c 17 87 3b  -...1"* @ . . . . .;
00d0  bd 48 db fd d0 d4 5f 1b  2d 79 4d b6 0c ae c7 ed  -H . . . . .-yM^
00e0  26 ba 7a e9 66 ce 4e 8d  28 d0 87 56 38 31 f8 1f  -& z f N ( . . . . .V81
00f0  69 46 92 41 e9 fc d4 b7  a3 09 9d b3 25 0f 67 5a  -iF A . . . . .% gZ
0100  49 47 e2 1c 53 71 15 34  0a 4c 1a 34 80 7a 6f 8e  -IG `Sq 4 L 4 zo

```

15.PROTOCOL ANALYSIS

15.1 DNS Analysis

Protocol: DNS (Domain Name System)

Purpose:

DNS converts domain names into IP addresses.

Observed Activities:

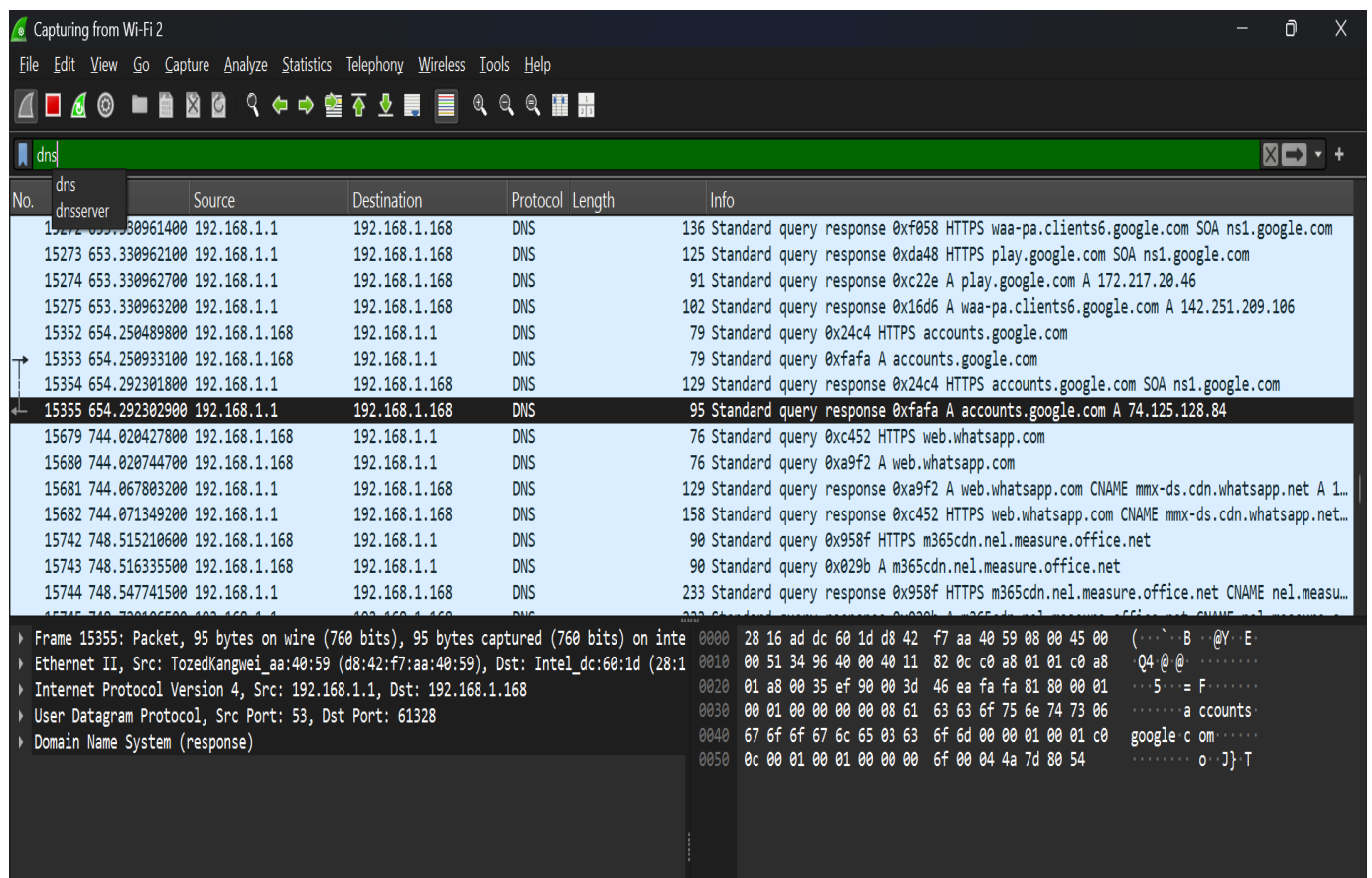
- DNS Queries • DNS Responses • Name Resolution

Cybersecurity Relevance:

DNS monitoring helps identify:

- Malicious domains • Command-and-Control servers • DNS tunneling attacks

DNS Protocol Analysis



Capturing from Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Source	Destination	Protocol	Length	Info
15272	192.168.1.1	192.168.1.168	DNS	136	Standard query response 0xf058 HTTPS waa-pa.clients6.google.com SOA ns1.google.com
15273	653.330962100	192.168.1.168	DNS	125	Standard query response 0xda48 HTTPS play.google.com SOA ns1.google.com
15274	653.330962700	192.168.1.168	DNS	91	Standard query response 0xc22e A play.google.com A 172.217.20.46
15275	653.330963200	192.168.1.168	DNS	102	Standard query response 0x16d6 A waa-pa.clients6.google.com A 142.251.209.106
15352	654.250489800	192.168.1.168	DNS	79	Standard query 0x24c4 HTTPS accounts.google.com
15353	654.250933100	192.168.1.168	DNS	79	Standard query 0xfafa A accounts.google.com
15354	654.292301800	192.168.1.168	DNS	129	Standard query response 0x24c4 HTTPS accounts.google.com SOA ns1.google.com
15355	654.292302900	192.168.1.168	DNS	95	Standard query response 0xfafa A accounts.google.com A 74.125.128.84
15679	744.020427800	192.168.1.168	DNS	76	Standard query 0xc452 HTTPS web.whatsapp.com
15680	744.020744700	192.168.1.168	DNS	76	Standard query 0xa9f2 A web.whatsapp.com
15681	744.067803200	192.168.1.168	DNS	129	Standard query response 0xa9f2 A web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net A 1...
15682	744.071349200	192.168.1.168	DNS	158	Standard query response 0xc452 HTTPS web.whatsapp.com CNAME mmx-ds.cdn.whatsapp.net...
15742	748.515210600	192.168.1.168	DNS	90	Standard query 0x958f HTTPS m365cdn.nel.measure.office.net
15743	748.516335500	192.168.1.168	DNS	90	Standard query 0x029b A m365cdn.nel.measure.office.net
15744	748.547741500	192.168.1.168	DNS	233	Standard query response 0x958f HTTPS m365cdn.nel.measure.office.net CNAME nel.measu...

Frame 15355: Packet, 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
Ethernet II, Src: TozedKangwei_aa:40:59 (d8:42:f7:aa:40:59), Dst: Intel_dc:60:1d (28:1d:67:6f:6f:6f)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.168
User Datagram Protocol, Src Port: 53, Dst Port: 61328
Domain Name System (response)

0000 28 16 ad dc 60 1d d8 42 f7 aa 40 59 08 00 45 00 (.....B...@Y..E..
0010 00 51 34 96 40 00 40 11 82 0c c0 a8 01 01 c0 a8 ..04.@@.....
0020 01 a8 00 35 ef 90 00 3d 46 ea fa fa 81 80 00 01 ...5...=F.....
0030 00 01 00 00 00 00 08 61 63 63 6f 75 6e 74 73 06a ccounts..
0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 01 00 01 c0 google c om.....
0050 0c 00 01 00 01 00 00 00 6f 00 04 4a 7d 80 54o..J}.T.....

15.2 ICMP Analysis

Protocol: ICMP (Internet Control Message Protocol)

Purpose:

Used for:

- Ping Requests • Ping Replies • Network Diagnostics

Observed Activities:

- Echo Requests • Echo Responses

Cybersecurity Relevance:

ICMP is useful for:

- Connectivity testing • Host discovery • Troubleshooting

ICMP Protocol Analysis

Capturing from Wi-Fi 2

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	icmp icmpv6	Source	Destination	Protocol	Length	Info
1193	17.467876200	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
1193	17.467876200	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x46c6, seq=0/0, ttl=64 (no response found!)
1777	38.442902700	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
1781	38.457671500	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x4775, seq=0/0, ttl=64 (no response found!)
8232	59.422901400	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
8235	59.438936600	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x482f, seq=0/0, ttl=64 (no response found!)
8406	80.413735900	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
8409	80.442401400	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x490a, seq=0/0, ttl=64 (no response found!)
8577	101.405065100	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
8580	101.433376400	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x4a50, seq=0/0, ttl=64 (no response found!)
8833	122.396436200	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
8836	122.424514400	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x4b09, seq=0/0, ttl=64 (no response found!)
8909	143.388121400	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)
8912	143.415491000	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x4bc2, seq=0/0, ttl=64 (no response found!)
8953	164.378913300	192.168.1.1	192.168.1.168	ICMP	60	Echo (ping) request id=0x090a, seq=0/0, ttl=64 (no response found!)

Frame 15639: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: TozedKangwei_aa:40:59 (d8:42:f7:aa:40:59), Dst: Intel_dc:60:1d (28:1d:60:1d:60:1d)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.168
Internet Control Message Protocol

0000 28 16 ad dc 60 1d d8 42 f7 aa 40 59 08 00 45 00 (.....B...@Y..E
0010 00 1c 32 8f 40 00 40 01 84 58 c0 a8 01 01 c0 a8 ..2@@X.....
0020 01 a8 08 00 8c 4e 6b b1 00 00 00 00 00 00 00 00NK.....
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00X.....

15.3 TCP Analysis

Protocol: TCP (Transmission Control Protocol)

Purpose:

Provides reliable communication between devices.

Observed Activities:

- TCP Handshake • ACK Packets • Established Connections

Cybersecurity Relevance:

TCP analysis assists in:

- Monitoring connections • Detecting anomalies • Investigating suspicious traffic

TCP Protocol Analysis

No.	Time	Source	Destination	Protocol	Length	Info
15212	642.045581900	192.168.1.168	51.116.253.170	TCP	55	[TCP Keep-Alive] 57036 → 443 [ACK] Seq=7634 Ack=9102 Win=65024 Len=1
15213	642.199133400	51.116.253.170	192.168.1.168	TCP	66	[TCP Keep-Alive ACK] 443 → 57036 [ACK] Seq=9102 Ack=7635 Win=4193024 Len=0 SLE=7634...
15249	645.199467100	44.226.92.154	192.168.1.168	TLSv1.3	79	Application Data
15250	645.201109600	192.168.1.168	44.226.92.154	TLSv1.3	83	Application Data
15251	645.479864800	44.226.92.154	192.168.1.168	TCP	60	443 → 59017 [ACK] Seq=1205 Ack=3213 Win=61440 Len=0
15261	651.491589900	192.168.1.168	74.248.73.245	TLSv1.2	105	Application Data
15262	651.651131500	74.248.73.245	192.168.1.168	TLSv1.2	94	Application Data
15263	651.705920900	192.168.1.168	74.248.73.245	TCP	54	55013 → 443 [ACK] Seq=868 Ack=681 Win=254 Len=0
15368	654.556984500	192.168.1.168	74.125.128.84	TCP	66	62242 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
15379	654.716657200	74.125.128.84	192.168.1.168	TCP	70	443 → 62242 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
15380	654.716772200	192.168.1.168	74.125.128.84	TCP	54	62242 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
15381	654.717468300	192.168.1.168	74.125.128.84	TCP	1466	62242 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 15382]
15382	654.717468300	192.168.1.168	74.125.128.84	TLSv1.3	403	Client Hello (SNI=accounts.google.com)
15393	654.875742900	74.125.128.84	192.168.1.168	TCP	64	443 → 62242 [ACK] Seq=1 Ack=1762 Win=268032 Len=0
15394	654.876994400	74.125.128.84	192.168.1.168	TLSv1.3	1470	Server Hello, Change Cipher Spec

Frame 15263: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0	Ethernet II, Src: Intel_dc:60:1d (28:16:ad:dc:60:1d), Dst: TozedKangwei_aa:40:59 (d8:4	Internet Protocol Version 4, Src: 192.168.1.168, Dst: 74.248.73.245	Transmission Control Protocol, Src Port: 55013, Dst Port: 443, Seq: 868, Ack: 681, Len
0000	d8 42 f7 aa 40 59 28 16	ad dc 60 1d 08 00 45 00	B...@Y(.....E
0010	00 28 f2 76 40 00 80 06	b1 1b c0 a8 01 a8 4a f8	(.v@.....J
0020	49 f5 d6 e5 01 bb 24 28	15 e5 f3 cd 58 23 50 10	I.....\$(.....X#P
0030	00 fe f8 f9 00 00		

16.SECURITY FINDINGS

The following protocols were observed:

• DNS • TCP • ICMP • UDP • TLS

The captured traffic demonstrated normal network communications.

No suspicious traffic was observed during the monitoring period.

Implemented Security Controls:

- ✓ Windows Defender Firewall Enabled
- ✓ WPA2 Wireless Security Enabled
- ✓ AES Encryption Enabled
- ✓ Wireshark Monitoring Implemented

17.TESTING AND VALIDATION

Validation Activities Completed:

- ✓ Wireshark Installed Successfully
- ✓ Npcap Installed Successfully
- ✓ Interface Detection Successful
- ✓ Live Traffic Capture Successful
- ✓ DNS Analysis Successful
- ✓ ICMP Analysis Successful
- ✓ TCP Analysis Successful
- ✓ Connectivity Testing Successful
- ✓ Firewall Verification Successful
- ✓ Wireless Security Verification Successful

18.CONCLUSION

This project successfully established a practical cybersecurity monitoring environment using Wireshark and Npcap. Live network traffic was captured and analyzed through the Wi-Fi 2 interface, enabling observation of real-world network communications.

The project demonstrated the ability to identify and analyze key network protocols including DNS, ICMP, TCP, UDP, and TLS. Network connectivity testing confirmed stable communication between devices, while ARP analysis provided visibility into local network addressing.

Security controls including Windows Defender Firewall and WPA2 wireless encryption were verified and documented. These controls contribute to protecting systems against unauthorized access and network-based threats.

The project provided valuable practical experience in packet capture, protocol analysis, network troubleshooting, security monitoring, and traffic inspection. These skills are essential for cybersecurity analysts, network administrators, SOC analysts, digital forensic investigators, and penetration testers responsible for protecting modern networks.

The objectives of the project were successfully achieved, and the implementation demonstrated the practical application of industry-standard tools for cybersecurity monitoring and network analysis.